

120 Days of Quantitative Finance

Month 3: Machine Learning and Deep Learning

Week 12: Deep Learning

Day 82: Sequence Modeling for Returns

1 Introduction

Financial markets generate time series data where observations are sequential and dependent. Returns at time t often depend on past returns, volatility, and market conditions.

Sequence modeling aims to capture these temporal dependencies using models such as:

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory (LSTM)
- Transformers

These models are designed to learn patterns across time rather than treating observations independently.

2 Return Definition

The return of an asset is defined as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

or in log form:

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right)$$

Returns are typically used instead of prices because they are more stable and easier to model.

3 Sequence Representation

Instead of modeling individual returns, we consider sequences:

$$(r_{t-n}, r_{t-n+1}, \dots, r_t)$$

The goal is to predict the next return:

$$\hat{r}_{t+1} = f(r_{t-n}, \dots, r_t)$$

This transforms the problem into a sequence-to-one prediction task.

4 Why Sequence Modeling is Important

Financial returns exhibit several temporal properties:

- Autocorrelation (short-term dependencies)
- Volatility clustering
- Regime changes

Traditional models often fail to capture these dynamics effectively.

Sequence models can learn:

- Patterns over time
- Nonlinear dependencies
- Long-term relationships

5 LSTM-Based Modeling

LSTM models process sequences step by step and maintain a memory state.

At each time step:

$$h_t = LSTM(r_t, h_{t-1})$$

The prediction is then:

$$\hat{r}_{t+1} = Wh_t + b$$

This allows the model to retain relevant historical information.

6 Transformer-Based Modeling

Transformers use attention mechanisms instead of recurrence.

The key idea is:

$$\hat{r}_{t+1} = f(r_{t-n}, \dots, r_t)$$

where the model assigns weights to different time steps using attention:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

This allows the model to focus on the most relevant past observations.

7 Applications in Finance

Sequence modeling is widely used in quantitative finance:

- Return prediction
- Volatility forecasting
- Regime detection
- High-frequency trading signals

For example, a model may learn that:

- Strong recent momentum increases future returns
- High volatility predicts risk
- Certain patterns precede market reversals

8 Challenges

Despite their power, sequence models face challenges:

- Financial data is noisy and non-stationary
- Signal-to-noise ratio is low
- Overfitting risk is high

Proper validation and regularization are essential.

9 Conclusion

Sequence modeling provides a powerful framework for understanding financial markets.

By modeling returns as sequences, deep learning models can capture temporal dependencies and complex patterns that traditional methods may miss.

These techniques are central to modern quantitative trading and alpha generation.